

CausalMedia-GH Teacher Dashboard

How Explanation Videos Affect Learning in Your School

NexusLearn Group 4 | KNUST | May 2026 | Supervisor: Dr Eric Osei

What This Report Is About

This report presents the causal effect of explanation videos on student learning across 10,000 students. It goes beyond simple correlations to estimate how much videos actually **cause** improvement for each student type, controlling for 12 background factors.

Key finding (v3 — updated): Rural students and those with slower internet connections show the **highest** benefit from videos (CATE ~+0.12). Urban and high-bandwidth students show a smaller positive effect. This suggests videos are most impactful as a supplementary resource where other learning materials are scarcer.

Key Findings at a Glance

Student Group	Video Effect	What This Means for You
All students (average)	+0.0392	Small positive effect overall
Q1: Weakest students	+0.055	Videos help — assign during revision
Q3: Above-average students	+0.031	Moderate benefit — continue assigning
Q4: Strongest students	+0.076	Highest benefit by achievement — keep up
Rural students	+0.124	Largest benefit — prioritise video access here
Low bandwidth (Category 1)	+0.126	Highest benefit — improve access urgently
Medium bandwidth (Cat 2)	-0.005	Near zero — investigate other factors
Urban / high bandwidth	+0.017	Modest benefit

Statistical note: Average effect = 0.0392 (95% CI: -0.603 to 0.682). Based on 10,000 students. CI includes zero, indicating findings are directional.

What Drives Each Student Result?

The chart below shows which variables push a student's video benefit up (right, red) or down (left, blue). Each dot is one student.

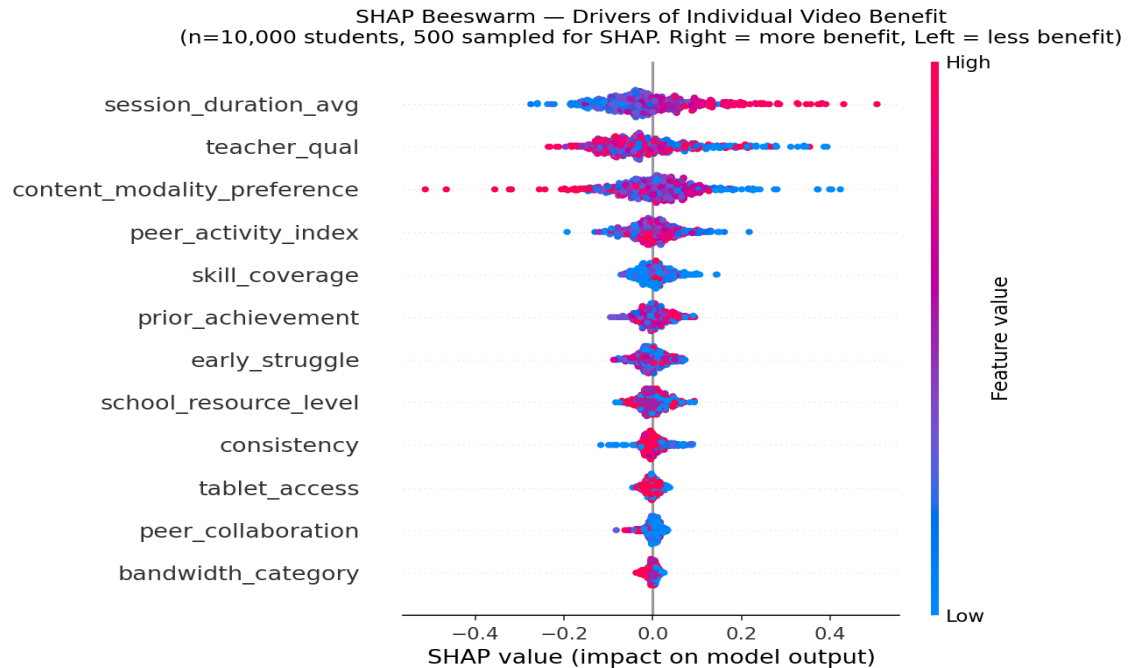


Figure 1: Global SHAP beeswarm — top variables by influence on video benefit.

Prior Achievement and Internet Quality Interaction

This chart shows how **prior achievement and bandwidth interact**. Low-bandwidth students (colour = 1) are spread across both positive and negative SHAP values, revealing that the effect is not uniform — suggesting other variables moderate the benefit within this group.

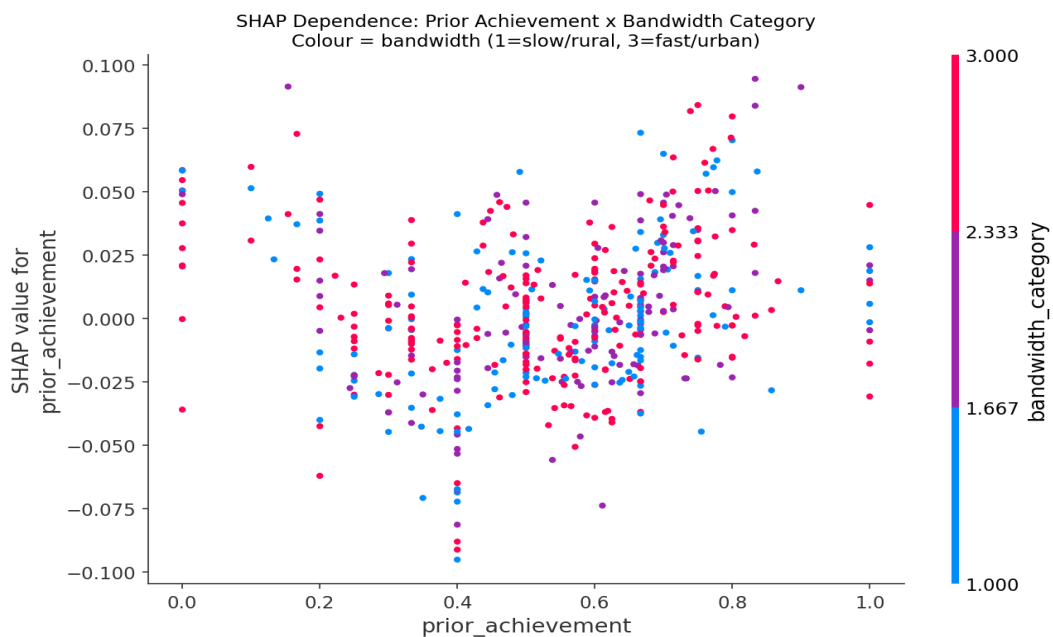
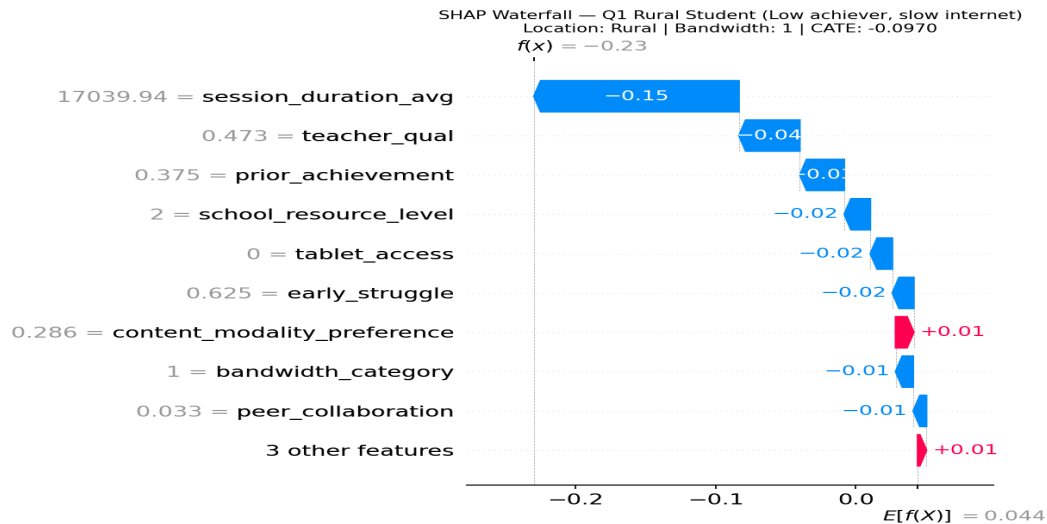


Figure 2: SHAP dependence plot — prior achievement x bandwidth category.

Three Student Profiles

Profile A — Rural Student, Low Prior Achievement

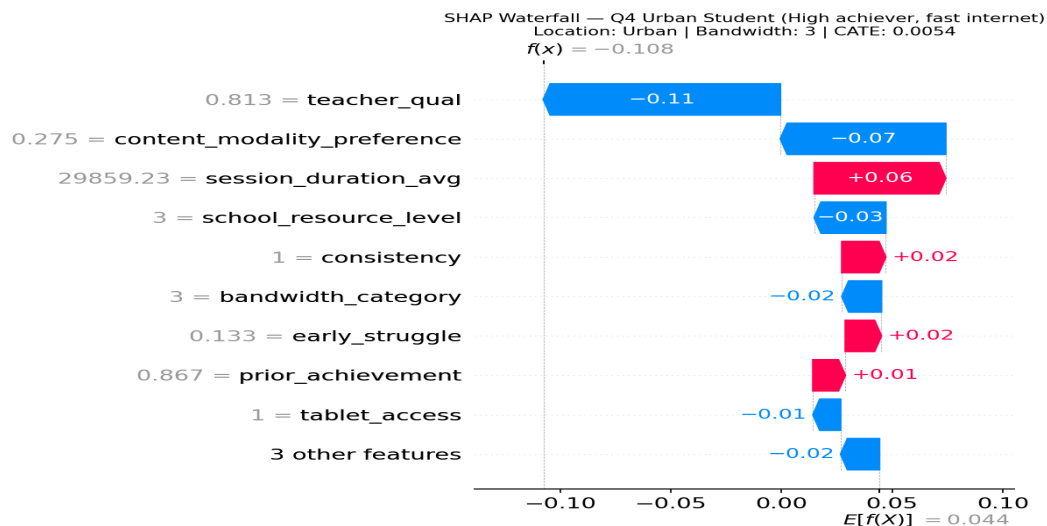
This student is in a rural area with slow internet and low prior achievement. According to the updated findings, rural low-bandwidth students show the highest causal benefit from videos. The SHAP waterfall shows which specific variables drive this result for this individual student.



Recommendation: **Ensure this student has video access.** Rural students benefit most — prioritise device and connectivity support for this group.

Profile B — Urban Student, High Prior Achievement

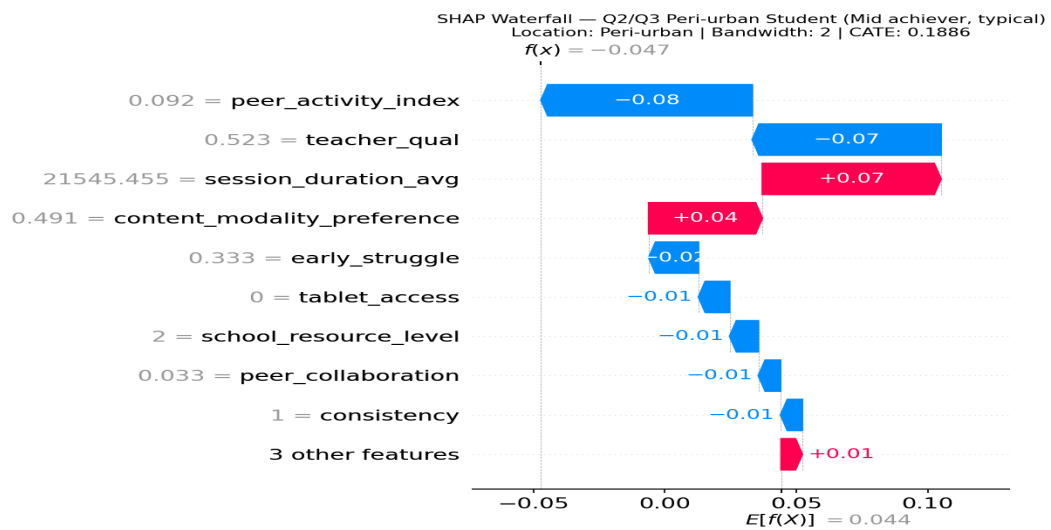
This urban student has fast internet and high prior achievement. The model shows a modest positive effect for this group. Urban high-achievers still benefit, but less dramatically than rural students.



Recommendation: **Continue video assignments** but also supplement with harder challenge materials. Videos alone may not stretch top students enough.

Profile C — Peri-urban Student, Mid Prior Achievement

This student represents the middle majority. Their bandwidth category (2) shows near-zero average effect, making this the most variable group. Individual monitoring is most important here.



Recommendation: **Monitor individually.** Track whether this student improves after video-heavy weeks. Adjust based on observed results.

5 Actions for Teachers

- **Prioritise video access for rural students.** They show the highest causal benefit. If your school is in a rural area, investing in device access and offline video downloads will have the most impact.
- **Investigate medium-bandwidth students individually.** Category 2 bandwidth shows near-zero average effect — meaning some students in this group benefit and some do not. Track their progress week by week.
- **Continue video assignments for urban and high-achieving students** — they still benefit modestly. Supplement with harder practice materials to keep top students challenged.
- **Focus on offline access in low-bandwidth areas.** Download videos in advance rather than relying on streaming. Slow internet does not reduce benefit — but it must not prevent students from watching the content at all.
- **Report unusual results to your IT coordinator.** If a student shows no improvement after consistent video engagement, there may be a device, connectivity, or engagement issue worth investigating.